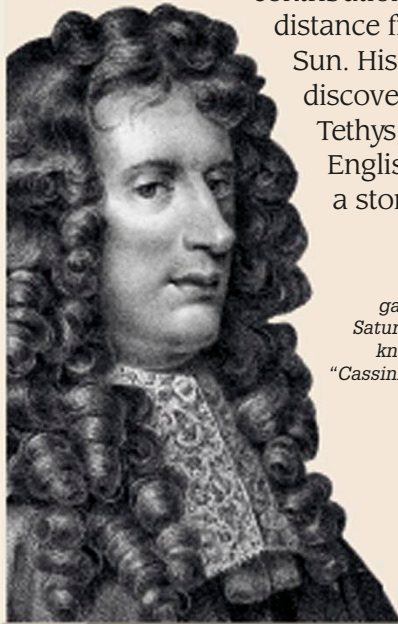
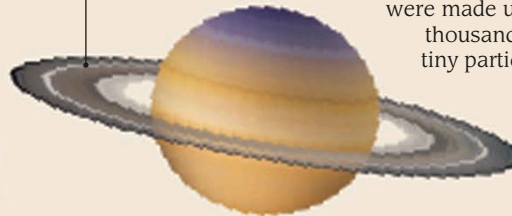


1625–1712 GIOVANNI CASSINI

Born in Italy, astronomer Giovanni Cassini moved to France in 1669, where he was put in charge of the Paris Observatory. Among his contributions to astronomy, Cassini calculated the distance from Earth to Mars, and from Mars to the Sun. His figures were close to current estimates. He discovered four of Saturn's moons—Iapetus, Rhea, Tethys, and Dione—and also shares credit with English scientist Robert Hooke for the discovery of a storm on Jupiter known as the Great Red Spot.



The large gap between Saturn's rings is known as the "Cassini Division."



Rings around Saturn

In 1675, Cassini identified a gap that appeared to divide Saturn's rings into two. He also correctly suggested that the rings were made up of thousands of tiny particles.

Isaac Newton was born in 1642, the year that Galileo died.



Dental hygiene

One of the first books on dentistry was published in 1685. It recommended brushing teeth only once a week. Not surprisingly, most people had rotten teeth, which had to be extracted by force.

1685

1690



Title page of John Ray's *Historia Plantarum*

1686

New term

In *Historia Plantarum*, a three-volume history of plants, English naturalist John Ray used the word "species" to describe a group of plants or animals sharing the same characteristics and able to breed with one another. It was the first biological use of the term, and established "species" as the basic unit of taxonomy (the classification of living things). His book described 18,600 species.

Isaac Newton
See pages 88–89

Newton's universal law of gravitation



Gravitation exerts an identical force on two objects, pulling them together.



Doubling the mass of both objects increases the forces to four times their original strength.



Doubling the distance between them reduces the forces by a quarter.

1687

Breakthrough science

Isaac Newton published a book called *Principia Mathematica* in which he described his three laws of motion and the universal law of gravitation. According to the latter, the force of gravity between two objects is stronger as their masses increase, and weaker when the distance between them is bigger. These four laws together form the basis of mechanics—the science of forces and how things move.

“One species never springs from the seed of another nor vice versa.”

John Ray, *Historia Plantarum*, 1686